ANNALS OF LABORATORY MEDICINE

Supplemental Data Table S1. Characteristics of the studies included

Reference	Country	N	Positive rate (%)	Reference test	Index test
[6]	Spain	979	12.0	CCNA	C. Diff Quik Chek Complete
		103	47.6		Xpert *C. difficile*
					GenomEra *C. difficile*
[7]	Canada, France	240	27.5	CCNA	*C. difficile* Tox A/B II
[8]	USA	601	13.3	CCNA	Lyra Direct *C. difficile*
		601	18.6	TC	
[9]	Switzerland	545	14.3	TC	C. Diff Chek-60
					LIAISON *C. difficile* GDH
					LIAISON *C. difficile* Toxins A&B
[10]	Australia	403	11.9	*C. difficile* culture	VIDAS *C. difficile* GDH
					C. Diff Quik Chek-60 GDH
[11]	China	416	26.9	*C. difficile* culture	VIDAS *C. difficile* GDH
[12]	UK, USA	1,914		*C. difficile* culture	C. Diff Quik Chek-60
			18.7	ChromID	VIDAS *C. difficile* GDH
			16.4	CCFA	
[13]	USA	211	15.6	TC	C. Diff Quik Chek Complete
					AmpliVue *C. difficile*
					BD MAX Cdiff
					Illumigene *C. difficile*
					Lyra Direct *C. difficile*
					Xpert *C. difficile* /Epi
[14]	Korea	302	19.5	TC	VIDAS *C. difficile* toxin A&B
					Illumigene *C. difficile*
[15]	Germany	199	14.1	TC	artus *C. difficile* QS-RGQ
					Xpert *C. difficile*
[16]	Denmark	299	12.7	TC	Illumigene *C. difficile*
					PCRFast *C. difficile*
					Xpert *C. difficile*
[17]	Sweden	419	16.2	*C. difficile* culture	C. Diff Quik Chek Complete
[18]	USA	351	13.1	TC	BD GeneOhm Cdiff
[19]	Korea	608	9.0	TC	C. Diff Quik Chek Complete
					VIDAS *C. difficile* toxin A&B
[20]	Korea	127	8.7	TC	VIDAS *C. difficile* toxin A&B
					AdvanSure CD
[21]	Japan	118	28.0	TC	C. Diff Quik Chek Complete
					X/Pect Toxin A/B
					Verigene CDF
[22]	Argentina	250	42.8	CCNA	C. Diff Quik Chek Complete
					RIDASCREEN *C. difficile* Toxin A/B

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Supplemental Data Table S1. Continued

Reference	Country	N	Positive rate (%)	Reference test	Index test
[23]	Austria	300	14.0	*C. difficile* culture	C. diff Chek-60
			11.7	TC	*C. difficile* Tox A/B II
			8.3	CCNA	LIAISON *C. difficile* GDH
					LIAISON *C. difficile* Toxins A&B
					Xpert *C. difficile*
[24]	Korea	258	21.3	TC	artus *C. difficile* QS-RGQ
					Xpert *C. difficile*
[25]	Korea	271	20.3	TC	VIDAS *C. difficile* GDH
					VIDAS *C. difficile* toxin A&B
					Xpert *C. difficile*
[26]	Japan	231	22.9	TC	BD MAX Cdiff
					C. Diff Quik Chek Complete
[27]	Germany	989	16.9	TC	AmpliVue *C. difficile*
					Illumigene *C. difficile*
[28]	Germany	500	14.8	*C. difficile* culture	RIDASCREEN GDH
			12.0	TC	RIDASCREEN *C. difficile* Toxin A/B
					RIDA QUIK
[29]	Israel	209	46.9	TC	AmpliVue *C. difficile*
					BDmax Cdiff
					GenomEra *C. difficile*
					Illumigene *C. difficile*
					Simplexa *C. difficile* Universal Direct
					Xpert *C. difficile*
[30]	USA	682	20.7	TC	Cobas Cdiff
[31]	Australia			TC	C. diff Chek-60
		334	7.2	Enrichment*	BD Max Cdiff
		332	6.6	Direct†	
[32]	South Africa	141	21.3	TC	ImmunoCard Toxins A & B
					VIDAS *C. difficile* Toxin A & B
					GenoType Cdiff
					Xpert *C. difficile*
[33]	Korea	191	16.8	TC	C. diff Quik Chek Complete
					Xpert *C. difficile*
[34]	Korea	350	30.9	*C. difficile* culture	VIDAS *C. difficile* GDH
[35]	Korea	339	25.7	TC	BD MAX Cdiff
					Illumigene *C. difficile*
					IMDx *C. difficile*
					Xpert *C. difficile*
[36]	Korea	243	28.8	TC	BD GeneOhm Cdiff
					Seeplex Diarrhea-B1 ACE detection

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Supplemental Data Table S1. Continued

Reference	Country	N	Positive rate (%)	Reference test	Index test
[37]	Korea	253	19.4	TC	VIDAS *C. difficile* toxin A&B
					Xpert *C. difficile*
[38]	Korea	555	21.8	TC	VIDAS *C. difficile* toxin A&B
[39]	Korea	1,596	22.7	TC	VIDAS *C. difficile* toxin A&B
[40]	Korea	228	51.3	TC	*C. difficile* Tox A/B II
					VIDAS *C. difficile* toxin A&B
[41]	Brazil	92	27.2	CCNA	*C. difficile* Tox A/B II
			25.0	TC	Remel ProSpecT *C. difficile* Toxin A/B
					RIDASCREEN *C. difficile* Toxin A/B
					Simplexa *C. difficile* Universal Direct
[42]	Korea	203	12.8	TC	AdvanSure CD
					Illumigene *C. difficile*
[43]	Japan	69	43.5	TC	C. Diff Quik Chek Complete
					Verigene CDF
[44]	Korea	254	34.3	TC	RIDASCREEN *C. difficile* Toxin A/B
					BD MAX Cdiff
					IMDx *C. difficile*
					Xpert *C. difficile*

*Enrichment in cooked meat broth supplemented with gentamicin, cefoxitin, and cycloserine was undertaken. Taurocholic acid was added to the enrichment broth as a germinant and, after incubation, broths were alcohol shocked and plated onto ChromID *C. difficile* agar; †Toxigenic culture was performed by direct plating onto ChromID *C. difficile* agar.
Abbreviations: CCNA, cell cytotoxicity neutralization assay; CCFA, cycloserine-cefoxitin fructose agar; TC, toxigenic culture.

Supplemental Data Table S2. Sensitivities and specificities of index tests

Index test	Reference	Sensitivity (%)	95% CI	Specificity (%)	95% CI
Compared with TC					
Well-type GDH EIA					
C. diff Chek-60	[9]	93.6	88.2–99.0	98.5	97.4–99.6
C. diff Chek-60	[23]	94.3	80.8–99.1	98.1	95.7–99.4
C. diff Chek-60	[31]	87.5	67.6–97.2	100.0	98.8–100.0
C. diff Chek-60	[31]	95.5	77.1–99.2	100.0	98.8–100.0
RIDA QUICK	[28]	81.7	69.6–90.5	91.1	88.1–93.6
RIDASCREEN GDH	[28]	88.3	77.4–95.2	91.1	88.1–93.6
Membrane-type GDH EIA					
C. Diff Quik Chek Complete	[13]	87.9	71.8–96.5	92.1	87.2–95.6
C. Diff Quik Chek Complete	[19]	91.0	85.4–96.6	92.4	90.1–94.7
C. Diff Quik Chek Complete	[21]	97.0	84.2–99.5	71.8	61.0–81.0
C. Diff Quik Chek Complete	[26]	92.5	81.8–97.9	94.4	89.9–97.3
C. Diff Quik Chek Complete	[33]	96.9	83.7–99.5	97.5	93.7–99.3
C. Diff Quik Chek Complete	[43]	93.3	77.9–99.0	92.3	79.1–98.3
Automated GDH EIA					
LIAISON *C. difficile* GDH	[9]	98.7	96.2–100.0	97.2	95.7–98.7
LIAISON *C. difficile* GDH	[23]	100.0	89.9–100.0	94.0	90.4–96.5
VIDAS *C. difficile* GDH	[25]	96.4	87.5–99.6	85.2	79.7–89.6
Well-type toxin AB EIA					
C. difficile Tox A/B II	[23]	54.3	36.7–71.2	97.7	95.1–99.2
C. difficile Tox A/B II	[40]	71.8	62.7–79.7	70.3	60.9–78.6
C. difficile Tox A/B II	[41]	65.2	44.9–81.2	98.5	91.9–99.7
Remel ProSpecT *C. difficile* Toxin A/B	[41]	60.9	40.8–77.8	91.3	82.3–96.0
RIDASCREEN *C. difficile* Toxin A/B	[28]	51.7	38.4–64.8	100.0	99.2–100.0
RIDASCREEN *C. difficile* Toxin A/B	[41]	60.9	40.8–77.8	97.1	90.0–99.2
RIDASCREEN *C. difficile* Toxin A/B	[44]	31.0	21.6–41.9	98.8	95.7–98.8
Membrane-type toxin AB EIA					
C. Diff Quik Chek Complete	[13]	21.2	9.0–38.9	98.9	96.0–99.8
C. Diff Quik Chek Complete	[19]	63.6	50.9–76.3	98.0	96.8–99.2
C. Diff Quik Chek Complete	[21]	45.5	28.1–63.6	94.1	86.8–98.1
C. Diff Quik Chek Complete	[26]	52.8	38.6–66.7	100.0	97.9–100.0
C. Diff Quik Chek Complete	[33]	46.9	29.1–65.3	99.4	96.5–99.9
ImmunoCard Toxins A and B	[32]	40.0	22.7–59.4	99.1	95.1–100.0
RIDA QUICK	[28]	55.0	41.6–67.9	99.1	97.7–99.8
X/Pect Toxin A/B	[21]	27.3	13.3–45.5	100.0	95.8–100.0
Automated toxin AB EIA					
LIAISON *C. difficile* Toxins A&B	[9]	60.0	51.6–68.4	92.3	85.1–99.6
LIAISON *C. difficile* Toxins A&B	[23]	68.6	50.7–83.1	95.5	92.2–97.6
VIDAS *C. difficile* toxin A&B	[14]	40.4	27.0–54.9	98.8	96.4–99.7
VIDAS *C. difficile* toxin A&B	[19]	75.5	63.5–87.5	97.4	96.1–98.7

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Supplemental Data Table S2. Continued

Index test	Reference	Sensitivity (%)	95% CI	Specificity (%)	95% CI
VIDAS *C. difficile* toxin A&B	[20]	63.6	30.9–88.9	100.0	96.8–100.0
VIDAS *C. difficile* toxin A&B	[25]	49.1	35.4–62.9	98.2	95.3–99.5
VIDAS *C. difficile* toxin A&B	[32]	50.0	31.3–68.7	99.1	95.1–100.0
VIDAS *C. difficile* toxin A&B	[37]	44.4	29.7–60.0	100.0	98.2–100.0
VIDAS *C. difficile* toxin A&B	[38]	69.2	59.5–77.7	96.7	94.5–98.2
VIDAS *C. difficile* toxin A&B	[39]	67.7	62.1–72.9	96.1	94.9–97.1
VIDAS *C. difficile* toxin A&B	[40]	68.5	59.0–77.0	79.2	70.0–86.6
NAAT					
AdvanSure CD	[20]	100.0	71.3–100.0	98.3	93.9–99.7
AdvanSure CD	[42]	84.6	65.1–95.6	98.3	95.1–99.6
AmpliVue *C. difficile*	[13]	93.9	79.7–99.1	98.9	96.0–99.8
AmpliVue *C. difficile*	[27]	91.0	85.6–94.9	89.1	86.7–91.1
AmpliVue *C. difficile*	[29]	92.9	85.8–97.1	100.0	96.7–100.0
artus *C. difficile* QS-RGQ	[15]	100.0	87.5–100.0	89.5	83.9–93.6
artus *C. difficile* QS-RGQ	[24]	98.2	90.2–99.7	93.6	89.3–96.5
BD GeneOhm Cdiff	[18]	97.8	87.0–99.8	97.7	95.1–98.9
BD GeneOhm Cdiff	[36]	95.7	88.0–99.1	96.5	92.6–98.7
BD MAX Cdiff	[13]	96.9	83.7–99.5	100.0	97.9–100.0
BD MAX Cdiff	[26]	98.1	89.9–100.0	98.9	96.0–99.9
BD MAX Cdiff	[29]	91.8	84.5–96.4	100.0	96.7–100.0
BD Max Cdiff	[31]	91.7	73.0–99.0	99.0	97.2–99.8
BD Max Cdiff	[31]	95.5	77.2–99.9	99.0	97.2–99.8
BD MAX Cdiff	[35]	88.5	79.9–94.3	89.3	84.8–92.8
BD MAX Cdiff	[44]	81.6	71.9–89.1	95.8	91.6–98.3
Cobas Cdiff	[30]	92.9	87.4–96.1	98.7	97.4–99.4
GenomEra *C. difficile*	[29]	91.8	84.5–96.4	99.1	95.1–99.9
GenoType Cdiff	[32]	86.7	69.3–96.2	88.3	80.8–93.6
Illumigene *C. difficile*	[14]	88.1	77.1–95.1	96.7	93.6–98.6
Illumigene *C. difficile*	[16]	94.7	82.2–99.2	95.8	92.6–97.9
Illumigene *C. difficile*	[27]	91.6	86.3–95.3	90.9	88.7–92.8
Illumigene *C. difficile*	[29]	92.9	85.8–97.1	100.0	96.7–100.0
Illumigene *C. difficile*	[35]	80.5	70.6–88.2	93.7	70.6–88.2
Illumigene *C. difficile*	[42]	92.3	74.8–98.8	99.4	96.9–99.9
IMDx *C. difficile*	[35]	85.1	75.8–91.8	94.4	90.9–96.9
IMDx *C. difficile*	[44]	62.1	51.0–72.3	99.4	96.7–99.9
Lyra Direct *C. difficile*	[8]	82.1	73.5–88.5	96.9	94.9–98.2
Lyra Direct *C. difficile*	[8]	89.3	81.7–94.1	98.8	97.2–99.5
Lyra Direct *C. difficile*	[8]	85.7	77.5–91.4	99.0	97.4–99.6
Lyra Direct *C. difficile*	[13]	87.9	71.8–96.5	98.9	96.0–99.8
PCRFast *C. difficile*	[16]	76.3	59.8–88.5	98.1	95.6–99.7
Seeplex Diarrhea-B1 ACE detection	[36]	90.0	80.5–95.9	97.1	93.4–99.0

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Supplemental Data Table S2. Continued

Index test	Reference	Sensitivity (%)	95% CI	Specificity (%)	95% CI
Simplexa *C. difficile* Universal Direct	[29]	94.9	88.5–98.3	100.0	96.7–100.0
Simplexa *C. difficile* Universal Direct	[41]	60.9	38.6–80.3	98.6	92.2–99.8
Verigene CDF	[21]	93.9	79.8–99.3	96.5	90.0–99.3
Verigene CDF	[43]	96.7	82.7–99.4	97.4	86.5–99.6
Xpert *C. difficile*	[13]	93.9	79.7–99.1	98.9	96.0–99.8
Xpert *C. difficile*	[15]	100.0	87.5–100.0	90.1	84.6–94.1
Xpert *C. difficile*	[16]	100.0	90.7–100.0	92.3	88.4–95.3
Xpert *C. difficile*	[24]	94.6	84.9–98.8	94.6	90.5–97.3
Xpert *C. difficile*	[25]	94.6	84.9–98.8	94.9	91.0–97.4
Xpert *C. difficile*	[29]	99.0	94.4–99.8	100.0	96.7–100.0
Xpert *C. difficile*	[32]	90.0	73.5–97.9	97.3	92.3–99.4
Xpert *C. difficile*	[33]	93.8	79.2–99.1	98.1	94.5–99.6
Xpert *C. difficile*	[35]	89.7	81.3–95.2	92.9	88.9–95.7
Xpert *C. difficile*	[37]	100.0	92.7–100.0	95.1	91.1–97.6
Xpert *C. difficile*	[44]	82.8	73.2–90.0	98.8	95.7–99.8
Compared with *C. difficile* culture					
Well-type GDH EIA					
C. diff Chek-60	[23]	88.1	73.6–95.5	99.6	97.5–100.0
RIDASCREEN GDH	[28]	87.8	78.2–94.3	93.7	90.9–95.8
Membrane-type GDH EIA					
C. Diff Quik Chek Complete	[17]	94.1	85.6–98.3	94.6	91.7–96.7
C. Diff Quik Chek-60	[10]	87.5	74.0–94.8	97.5	95.1–98.7
C. Diff Quik Chek-60	[12]	93.0	89.8–95.4	96.7	95.6–97.5
C. Diff Quik Chek-60	[12]	96.8	94.2–98.5	94.9	93.8–96.0
Automated GDH EIA					
LIAISON *C. difficile* GDH	[23]	100.0	89.6–100.0	96.5	93.3–98.3
VIDAS *C. difficile* GDH	[10]	95.8	85.7–99.4	91.6	88.3–94.4
VIDAS *C. difficile* GDH	[11]	100.0	96.8–100.0	92.8	89.3–95.4
VIDAS *C. difficile* GDH	[12]	93.0	89.8–95.4	91.8	90.3–93.1
VIDAS *C. difficile* GDH	[12]	95.9	93.0–97.8	90.0	88.4–91.4
VIDAS *C. difficile* GDH	[34]	97.2	92.1–99.4	87.2	82.3–91.1

Abbreviations: TC, toxigenic culture; CI, confidence interval; GDH, glutamate dehydrogenase; EIA, enzyme immunoassay; NAAT, nucleic acid amplification test.

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